

BRAC University
Department of Electrical and Electronic Engineering
EEE/ECE-103IL: Computer Programming Laboratory
Summer 2026 (Section: 02)

Lab Final

1. Develop a **C program** that dynamically allocates memory to store **N temperature readings** collected every hour by a data logger. The value of **N** will be provided by the user, and the temperature readings will also be entered by the user. Due to random sampling, the temperature readings are stored in no particular order. [CO4] (10)

For proper analysis:

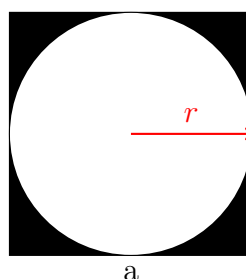
- The reading corresponding to the **coldest hour must appear first**
- The reading corresponding to the **hottest hour must appear last**
- This arrangement helps engineers observe the gradual temperature rise throughout the day

Write a C program that performs the following tasks:

- Dynamically allocate memory to store N temperature readings
- Read **N** temperature values from the user
- Modify the stored readings to satisfy the analysis requirement
- Display the final sequence of temperature readings

Constraints:

- Dynamic memory allocation is mandatory
 - Do not use any built-in functions
2. The diagram below shows a circle inscribed in a square where the center of the square and the center of the circle coincide. [CO4] (8)



Assume that a is the side length of the square (user-defined input). Write a C program to:

- Calculate the area of the square
- Calculate the area of the inscribed circle
- Determine how many times the area of the square is greater than the area of the circle and print it upto two decimal places

Use the formula:

$$\text{Area of circle} = \pi r^2$$

3. The following program is supposed to print **all integers** that are not divisible by 3 and 5 and lie between 1 and a user-defined number (inclusive). The program also counts the total **number** of such integers and prints the result. [CO4] (7)

No.	Code	Explanation/Corrected
1		
2	int main() {	
3	int end;	
4	printf("Enter the end number: ");	
5	scanf("%f", &end);	
6	int i, Count;	
7	count = 0;	
8	for(i=1; i<end; i++) {	
9	if ((i/3 != 0) & (i/5 != 0)) {	
10	count = count + 1;	
11	printf("%d\n", i)	
12	}	
13		
14	printf("Total count is: %c\n", Count);	
15	}	